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CSE-330

Practice-Script: 1

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# Practice Problems: Chapter 1

1. The floating point representation can be expressed in any of the following forms:

| | | |
|---|---|---|
| Standard Normalized Form | : | $F = \left(\pm\ 0.d_1d_2d_3\cdots d_m\right)_\beta \times \beta^e;\quad d_1 \neq 0.$ |
| IEEE Normalized Form | : | $F = \left(\pm\ 0.1d_1d_2d_3\cdots d_m\right)_\beta \times \beta^e.$ |
| IEEE Denormalized Form | : | $F = \left(\pm\ 1.d_1d_2d_3\cdots d_m\right)_\beta \times \beta^e.$ |

a) Consider a system with $\beta = 2, m = 4$, and $-3 \leq e \leq 4$. Find out the **maximum** and **minimum** numbers this system can store with and without **negative support**. Express the numbers both in binary and decimal digits for all three forms.

b) How many numbers can this system represent or store in all these forms?

c) Using **Standard Form**, find all the decimal numbers without **negative support**, plot them on a real line, and show if the number line is **equally spaced** or not.

d) For the **IEEE standard** for double-precision (64-bit) arithmetic, find the **smallest positive number** and the **largest number** representable by a system that follows this standard. Do not find their decimal values, but simply represent the numbers in the following format:

$$\left(\pm\ 0.1d_1d_2d_3\cdots d_m\right)_\beta \times \beta^{e-exponentBias}.$$

Be mindful of the conditions for representing $\pm\infty$ and $\pm 0$ in this IEEE standard.

e) In the above IEEE standard, if the exponent bias were to be altered to $exponent\ Bias = 500$, what would the **smallest positive number** and the **largest number** be? Write your answers in the same format as in part (d). Note that the conditions for representing $\pm\infty$ and $\pm 0$ are still maintained as before.

2. If $x = 3/8$ and $y = 5/8$, find $fl(x \times y)$ where $m = 4$. Also check whether $x \times y = fl(x \times y)$. If not, find the **rounding error** of the product of these two numbers.
3. Consider the quadratic equation, $x^2 - 60x + 1 = 0$. Working to **6 significant figures**, compute the **roots** of the quadratic equation and check that there is a **loss of significance**. Find the **correct roots** such that loss of significance does not occur.
4. Given $\beta = 2,\ m = 5,\ -100 \leq e \leq 100$. Using the IEEE **Normalized form**, answer the following:

   a) Compute the Machine Epsilon ($\epsilon_M$).

   b) Compute the minimum of $|x|$.

   c) How many non-negative numbers can you represent using this system?

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5. Consider the quadratic equation $x^2 - 16x + 3 = 0$. Explain how the loss of significance occurs in finding the roots of the quadratic equation if we restrict to **4 significant figures**. Discuss how to avoid this and find the roots.

6. Given a system parameterized by $\beta = 2$, $m = 3$, and $e_{min} = -1 \leq e \leq e_{max} = 2$, where $e \in Z$. For this system answer the following:

   (a) Find the floating-point representation of the numbers $(6.25)_{10}$ and $(6.875)_{10}$ in the Normalized Form. That is, find $fl(6.25)_{10}$ and $fl(6.875)_{10}$.

   (b) What are the rounding errors $\delta_1, \delta_2$ in part (a)?

   (c) Can the values $(6.25)_{10}$ and $(6.875)_{10}$ be represented in the Denormalized Form? If so, find the floating-point representations. If not, then concisely explain why?

   (d) Find the rounding error for Standard Form, Normalized and Denormalized Form.

7. Consider the **real number x =** $(8.235)_{10}$

   (a) First convert the decimal number x in binary format at least up to 8 binary places.

   (b) What will be the binary value of x [Find fl(x)] if you store it in a system with **m = 6** using the **Denormalized** form of floating point representation.

   (c) Now convert back to decimal form the stored values you obtained in the previous part, and calculate the **rounding error of both numbers**.

8. Consider the quadratic equation:

$$x^2 - 12x + 5 = 0$$

   a) Compute the roots of the quadratic equation while keeping to **four significant figures**.

   b) Explain how **loss of significance** occurs in this case due to the subtraction of nearly equal numbers.

   c) Discuss an alternative approach to computing the roots to **avoid loss of significance**, and use this method to determine the correct roots.

9. Consider a computing system with base $\beta = 2$, $m = 3$, and $e_{min} = -3 \leq e \leq e_{max} = 2$

   a) In the Standard form of this system, determine the total number of representable values including support for negative numbers. Also, compute the maximum value of delta.

   b) Express the floating-point representations (binary format) for the numbers $x = 4/8$ and $y = 7/8$ in this system.

   c) Compute $fl(x \times y)$ and determine whether this value can be stored within the given floating-point system.

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1) a) $\beta = 2$, $m = 4$

$e_{min} = -3$

$e_{max} = 4$

$e = \{-3, -2, -1, 0, 1, 2, 3, 4\}$

$(0.1111)_2 = 0.9375$

$(0.1000)_2 = 0.5$

## Standard Normalized Form

Maximum number $= +(0.1111)_2 \times 2^4 = 15$

Minimum number $= +(0.1000)_2 \times 2^{-3} = 0.0625$

## IEEE Normalized Form

Maximum number $= (0.1111)_2 \times 2^4 = 15$

Minimum number $= (0.1000)_2 \times 2^{-3} = 0.0625$

## IEEE Denormalized Form

Maximum number $= (1.1111)_2 \times 2^4 = 15$

Minimum number $= (1.0000)_2 \times 2^{-3} = 0.125$

b) Total amount of numbers in standard normalized forms,

$N = (2^3 \times 8) \times 2$

$= (8 \times 8) \times 2$

$N = 128$

Total amount of numbers in IEEE Normalized form,

$N = (2^4 \times 8) \times 2$

$= (16 \times 8) \times 2$

$= 256$

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b) Total amount of numbers in IEEE,

$N = (2^4 \times 8) \times 2$

$N = 256$

c) SKIP

d) In 64-bit architecture,

Range of $e = [-1022, 1025]$

Highest positive number = $(0.111111\ldots1_{52})_2 \times 2^{1024}$

Lowest positive number = $(0.10000\ldots0_{52})_2 \times 2^{-1021}$

e) P

$x = \frac{3}{8}$

$y = \frac{5}{8}$

$x \times y = 0.234375$

$fl(x \times y) = 0.2344$

$\therefore (x \times y) \neq fl(x \times y)$

Rounding error = $0.234375 - 0.234400$

$= (0.000025)$

Ans to the Qno: 2

$x = \frac{3}{8}$

$y = \frac{5}{8}$

$x \times y = (0.23975)_{10}$

$= (0.001111)_2$

$fl(x \times y) = (0.0011)_2 \times 2^0$

$fl(x \times y) \neq (x \times y)$

Rounding error = $0.001111 - 0.0011$

$= 0.000011$

.23975 x 2 = 0.46875
.46875 x 2 = 0.9375
.9375 x 2 = 1.875
.875 x 2 = 1.75
.75 x 2 = 1.5
.5 x 2 = 1.0

Ans to the Qno: 3

$x^2 - 60x + 1 = 0$

$x = \frac{+60 \pm \sqrt{(-60)^2 - 4}}{2}$

$x = \frac{60 \pm \sqrt{3596}}{2}$

$x = \frac{60 \pm \sqrt{4 \times 899}}{2}$

$= \frac{60 \pm 2\sqrt{899}}{2}$

$= 30 \pm \sqrt{899}$

$x_1 = 59.9833$

$x_2 = 0.016713$

$\sqrt{899} = 29.9833$

$x_1 = x_p \pm \delta_1 = 30 + 29.9833 = 59.9833$

$x_2 = x - \delta_1 = 30 - 29.9833$

$= 0.0167$

∴ There is a loss of significance

$\alpha + \beta = \frac{-b}{a}$

$\alpha\beta = ca$

$\alpha + \beta = +60$

$\alpha\beta = 1$

$\beta = \frac{1}{\alpha}$

$\alpha + \frac{1}{\alpha} = +60$

$\alpha^2 + 1 = +60\alpha$

$\alpha = 30 + \sqrt{899}$

$\beta = 30 - \sqrt{899}$

## Ans to the Q no: 4

a) $\in =$

b) $x = (0.100000)_2 \times e^{-100}$

c) No of non-negative numbers $= (2^5 \times (100+100+1))$

$= 6432$

## Ans to the Q no: 5

$x^2 - 16x + 3$

$x = \frac{16 \pm \sqrt{(-16)^2 - 4(3)}}{2}$

$x = \frac{16 \pm \sqrt{256 - 12}}{2}$

$x = \frac{16 \pm \sqrt{244}}{2}$

$x = \frac{16 \pm \sqrt{4 \times 61}}{2}$

$x_{\mp} = \frac{16 \pm 2\sqrt{61}}{2}$

$x = 8 \pm \sqrt{61}$

$x_1 = 8 + \sqrt{61} = 15.81$

$x_2 = 8 - \sqrt{61} = 0.1898$

$\sqrt{61} = 7.810$

$x_1 = 8 + 7.810 = 15.81$

$x_2 = 8 - 7.810 = 0.19$ — (significance loss)

$\alpha + \beta = \frac{-b}{a}$

$\alpha + \beta = 16$

$\alpha\beta = 3$

$\alpha = 15.81$

$\alpha + \beta = 16$

$\alpha\beta = 3$

$\beta = \frac{3}{\alpha}$

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## Ans to the Qno: 6

a) $(6.25)_{10} = (110.01)_2$

$.25 \times 2 = 0.5$
$0.5 \times 2 = 1.0$

$(6.875)_{10} = (110.111)_2$

$.875 \times 2 = 1.75$
$.75 \times 2 = 1.5$
$.5 \times 2 = 1.0$

$fl(6.25)_{10} = (0.1100) \times 2^2$

## Ans to the Qno: 7

$.325 \times 2 = 0.65$
$.65 \times 2 = 1.30$
$.30 \times 2 = 0.60$
$.60 \times 2 = 1.20$
$.20 \times 2 = 0.40$
$.40 \times 2 = 0.80$
$.80 \times 2 = 1.60$
$.60 \times 2 = 1.20$
$.20 \times 2 = 0.40$

a) $x = (8.325)_{10}$

$x = (1000.0101)_2$

b) $fl(x) = (1.0000101)_2 \times 2^3$

$= (1.000010)_2 \times 2^3$

~~c) $(1.000010)_2 = (1.03125)_{10}$~~

c) $(1000.010)_2 = (8.25)_{10}$

$\delta = 8.325 - 8.25$

$\delta = 0.075$

# Ans to the Q no: 8

a) $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$= \frac{12 \pm \sqrt{12^2-4(1)(5)}}{2}$

$= \frac{12 \pm \sqrt{144-20}}{2}$

$= \frac{12 \pm \sqrt{124}}{2}$

$= \frac{12 \pm \sqrt{4 \times 31}}{2}$

$= \frac{12 \pm 2\sqrt{31}}{2}$

$= 6 \pm \sqrt{31}$

$x_1 = 11.57$ and $x_2 = 0.4322$

b) $\sqrt{31} = 5.56776 \approx 5.568$

$x_1 = 6 + 5.56776$ and $x_2 = 6 - 5.56776$

$x_1 = 11.56776$ and $x_2 = 0.432$ (loss of significance)

c) $\alpha + \beta = \frac{-b}{a}$

$\alpha + \beta = 12$

$\alpha\beta = 5$

$\alpha = 11.56776$

$\beta = 12 - 11.56776$

$\beta = 0.4322$

# Ans to the Qno: 9

$\beta = 2, m = 3$

$e_{min} = -3$

$e_{max} = 2$

a) Total no of representable values = $(2^{3-1} \times (3+1) \times 2)$

$= 48$

(0 to 1)

$(0.100)_2$ $(0.100)$

Max value of delta = $0.1010 - 0.1005 = (0.0005)_{ans}$

$\cdot 5 \times 2 = 1.0$

$\cdot 875 \times 2 = 1.75$

$\cdot 75 \times 2 = 1.5$

$\cdot 5 \times 2 = 1.0$

$\cdot 4375 \times 2 = 0.875$

$\cdot 875 \times 2 = 1.75$

$\cdot 75 \times 2 = 1.5$

$\cdot 5 \times 2 = 1.0$

b) $x = (0.5)_{10} \Rightarrow (0.1)_2$

$y = (0.875)_{10} \Rightarrow (0.111)_2$

$fl(x) = (0.1)_2 \times 2^0$

$fl(y) = (0.111)_2 \times 2^0$

c) $x \times y = (0.4375)_{10}$

$= (0.111)_2$

~~$\therefore fl(x)$~~

$\therefore fl(x \times y) = (0.111)_2 \times 2^0$